Danial Jafarzadeh Jazi

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LinkedIn: LinkedIn | GitHub: GitHub | Nationality: Iranian

Research Interests

Aspiring AI systems researcher at the intersection of efficient deep learning and computational neuroscience. I develop GPU-optimized kernels and tensor computing frameworks to accelerate multimodal transformer architectures, with application to brain state decoding from neuroimaging data (MRI, EEG). Seeking a research-intensive postgraduate program to advance scalable neural and medical AI systems.

Education

- **Islamic Azad University – Central Tehran Branch** *Sep. 2021 – Sep. 2025 (graduated)*
  B.Sc. in Computer Science Tehran, Iran
  - Cumulative GPA: 17.65/20.00 (approximately 3.5/4.0)
  - **Upper-division / Major GPA (last 94 credits):** 18.74/20.00 (approximately 3.75/4.0)
  - Relevant coursework: Machine Learning, Artificial Intelligence, Data Structures & Algorithms, Linear Algebra, Probability & Statistics

Research Experience

- **Islamic Azad University** *Sep. 2023 – Sep. 2024*
  Research Assistant Tehran, Iran
  - **Multimodal neuroimaging (first author):** Proposed a transformer-based framework for brain state decoding from structural MRI data, designing multi-head cross-modal attention mechanisms and positional encoding strategies in PyTorch. Currently implementing the full training pipeline, with core model components under active development. First-author manuscript describing the proposed architecture available on arXiv.
  - **Medical image analysis:** Designed and trained CNN-based architectures for brain tumor detection and classification on large-scale MRI datasets, handling DICOM volume ingestion, resampling, normalization, and data augmentation protocols.
  - **Mentorship:** Supervised and mentored 3 undergraduate researchers on machine learning experimental design, code review, and reproducible research practices within the lab.

Teaching Experience

- **Islamic Azad University** *Jan. 2025 – Sep. 2025*
  Teaching Assistant – Advanced Programming (Supervisor: Dr. Elham Taghizadeh) Tehran, Iran
  - Supervised students covering object-oriented design, data structures, and software engineering principles in Python
  - Conducted weekly problem-solving sessions, facilitated code reviews, and assisted in assignment design and grading

Manuscript In Preparation

[P.1] D. Jafarzadeh Jazi and M. Hajiesmaeili, "Transformers for Multimodal Brain State Decoding," *arXiv preprint*, 2025. [arXiv:2512.08462]

Selected Projects

- **GPT Language Model from Scratch** *2025*
  Python, PyTorch, tiktoken
  - Built a complete GPT-2-style decoder-only transformer (∼124M parameters, 12 layers, 12 heads) from scratch in PyTorch with multi-head causal self-attention, learned positional embeddings, and GELU-activated feed-forward blocks. Developed end-to-end tokenization, training, and inference scripts.
  - Implemented full training pipeline: AdamW with cosine LR schedule and warmup, gradient clipping, mixed-precision (BF16), and periodic checkpointing; achieved validation perplexity ∼3.0 on Shakespeare with coherent autoregressive generation via temperature and top-k sampling. repository at github.com/Danialjfz/GPT-Language-Model
- **GPU Performance Engineering: Opti-GEMM** *2025 — Ongoing*
  C++, CUDA, CMake, Nsight Compute

- Systematic optimization of General Matrix Multiply (GEMM) kernels from naive global-memory implementation through shared-memory tiling, register blocking, and warp-level primitives
- Empirical cross-architecture benchmarking on Tesla T4 (Turing) and P100 (Pascal), documenting 5× speedup via manual SMEM tiling on cache-less Pascal vs. hardware-cached Turing
- Profiling with Nsight Compute against cuBLAS baselines; repository at github.com/Danialjfz/Opti-GEMM

Technical Skills

Programming:	Python, C/C++, CUDA, R, SQL
ML & Data Science:	PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn
Tools & Platforms:	Git/GitHub, LaTeX, Mendeley, Linux
Mathematics:	Linear Algebra, Probability & Statistics, Optimisation, Operations Research

Language Proficiency

Persian: Native | **English:** IELTS Academic 7.5 (L: 8.0 R: 8.5 W: 6.5 S: 7.0), August 2025 | **German:** A1–A2

Certifications & Memberships

Elements of AI	University of Helsinki & MinnaLearn	Nov. 2023
Writing in Plain Language	LinkedIn Learning	Oct. 2024

International Association of Engineers (IAENG), Membership ID: 530744 — *August 2025 – Present*

References

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